Evaluation of TurtleBot 4 Platform for Autonomous Stock-Checking Applications: A Sensor Characterization Report

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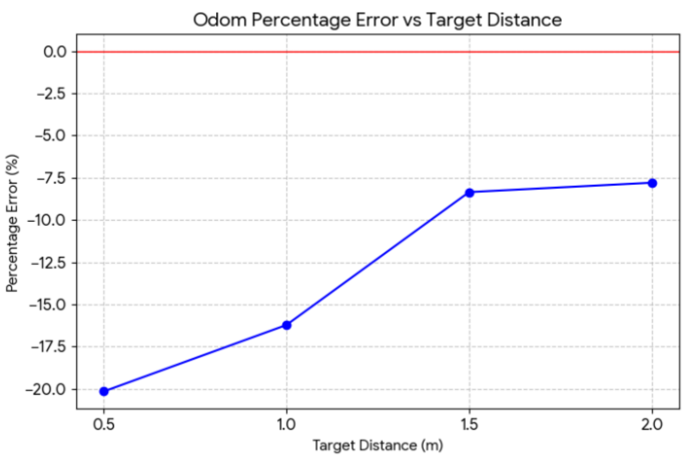
Abstract
This report evaluates the suitability of the TurtleBot 4 as a candidate platform for a real-world autonomous robot application. The investigation focuses on characterizing the performance of the wheel odometry, 2D LiDAR, and camera in an environment without external positioning infrastructure. Quantitative analysis of 20 linear trials and square path tests reveals a significant systematic under-reporting bias in the odometry and a closing error of 0.090m. While the 2D LiDAR demonstrated high precision, systematic range offsets were observed. Furthermore, the visual detection system proved highly sensitive to ambient lighting and material properties. Consequently, the TurtleBot 4 is not recommended for purchase in its current state without the implementation of specific technical enhancements.

1. Investigation A1: Linear Displacement Accuracy

1.1 Linear Displacement Experimental Results
This investigation aims to quantify the accuracy and repeatability of the TurtleBot 4 odometry during linear motion by conducting multiple repetitive trials across four distance gradients: 0.5m, 1.0m, 1.5m, and 2.0m. For detailed raw data regarding 20 trials, please refer to Appendix A.

1.2 Data Interpretation
In-depth analysis of the 20 experimental datasets reveals three key engineering characteristics regarding the sensor's performance.
Consistent Systematic Bias: The data I collect indicates that the percentage error for all experimental groups is negative, demonstrating that the odometry consistently under-reports the actual travel distance. In practice, this means that when the robot is commanded to travel a fixed distance, it fails to reach the physical target point. Error Convergence with Increased Distance: As illustrated in the figure below, the error exhibits a distinct non-linear trend. At short distances (0.5m), the percentage error is as high as -20.15%, whereas at a distance of 2.0m, the error converges to -7.80%. This suggests the presence of a significant fixed loss component—potentially caused by instantaneous wheel slippage during startup or initial encoder pulse loss. As the travel distance increases, the relative contribution of this initial cumulative error to the total distance decreases.



Based on 20 sets of experimental data, the precise percentage error was calculated, and a corresponding trend line graph was generated. It can be clearly observed that the line trends upward to the right, demonstrating that as the distance increases, the percentage error decreases.

2. Investigation A2: Closed-Loop Drift

2.1 Summary of Experimental Results
In this experiment, the robot was required to perform four 90-degree turns and travel along four straight segments. By recording the robot's final position after completing the path, the accumulated odometry error can be evaluated. Detailed data can be found in Appendix D.

2.2 Data Interpretation
Significant Offset: The average final position is (-0.084, 0.026), indicating that after completing the loop, the robot stopped approximately 9 cm to the left rear of the starting point.
Accumulation of Rotational Error: According to the data, the final heading angle does not return to zero but instead shows a deviation of approximately 1° to 4°. This suggests that during the four 90-degree turns, factors such as wheel slippage or inaccurate rotation parameters prevented a perfect 360-degree rotation. Even a small error of 1° per turn can accumulate over four segments, resulting in a noticeable positional drift.

3. Investigation B1: Range Accuracy at Known Distances

3.1 LiDAR Range Accuracy Test
In this experiment, the robot was positioned at known distances (0.5 m, 1.0 m, and 2.0 m) from a flat wall or obstacle. A total of 50 samples were collected at each distance to evaluate measurement accuracy and statistical noise. A summary of the ranging performance statistics for this test can be found in Appendix B.

3.2 Data Interpretation
High Measurement Precision (Low Noise): A notable characteristic is the extremely small standard deviation. This indicates that the LiDAR measurements are highly stable, with minimal random noise, and all readings are tightly clustered within a very narrow range.
Systematic Positive Bias: In contrast to the underestimation observed in odometry, the LiDAR consistently shows an overestimation bias. The measured distances are slightly larger than the ground truth (e.g., approximately 1.167 m measured at a true distance of 1.0 m).
Engineering Explanation: This type of error is typically not due to sensor malfunction but arises from coordinate frame misalignment. The LiDAR reports distances from its own sensor origin; if there is an uncompensated physical offset between the LiDAR center (you can see this picture below) and the robot's reference frame (e.g., rotation center), a systematic bias will be introduced.

4. Investigation B2: Environment Snapshot Analysis

4.1 Experimental Observations

Using the 360° environment scan data obtained from *lidar_logger.py*, we can clearly identify the boundaries of the laboratory and the internal obstacles.

Geometric Reconstruction Capability: From the PNG outputs, the LiDAR successfully reconstructs the rectangular layout of the room. The wall boundaries appear as straight and well-defined lines, demonstrating the sensor's high angular resolution in the horizontal plane.

Obstacle Detection: The scattered points in the scan represent objects such as table legs, supports, or storage racks within the lab. In *snapshot3.png* (Appendices E), a clear shadowing effect can be observed—where closer obstacles block the laser beams from reaching objects behind them. In a stock-checking scenario, this implies that the robot must reposition itself from multiple viewpoints to achieve complete coverage of inventory shelves.

Data Completeness and Blind Spots: The distribution of scan data is highly uniform across all directions, with no significant blind spots. Most detected points fall within the range of 0.15 m to 3 m, which fully covers the sensing requirements for navigation in warehouse aisles.

4.2 Data Interpretation

Robustness to Noise: There are almost no randomly scattered isolated points (i.e., outliers or noise) visible in the scan. This is consistent with the extremely low standard deviation observed in the B1 experiment, indicating that the LiDAR provides highly reliable data in a static environment.

Validation of Physical Offset: In the snapshot, the distance distribution from the robot center (coordinate origin) to the surrounding walls appears highly regular. When combined with the systematic positive bias identified in B1 (approximately +15 cm overestimation), it can be inferred that this bias is consistent in all directions.

5. Investigation C2: Red Object Detection

5.1 Visual Marker Detection Test

This experiment evaluates the ability of the OAK-D camera to detect red inventory markers at different distances. Using colour thresholding techniques, the number of detected red pixels (Red Pixel Count) is quantified to assess detection performance.

Results Table C2： Detection Observations

Distance (m)	Surface / Object	Red pixels detected	Object visible? (Y/N)
0.5m	Red Object	16654	Y
1.0m	Red Object	13599	Y
1.5m	Red Object	8541	Y
2.0m	Red Object	5135	Y
3.0m	Red Object	3399	Y

5.2 Critical Environmental Constraints

During testing, it was observed that the reliability of the vision system is highly dependent on external environmental conditions.

Lighting Sensitivity: The experiment shows that the camera can only maintain stable pixel detection under sufficient lighting conditions. In low-light environments, increased sensor noise and color distortion cause

significant variation in the HSV threshold range for red, resulting in a reduced detection frequency (Hz) or even complete loss of the target. Refer to Appendix C for details.
Material Properties: The performance of the vision system is strongly influenced by the physical properties of red objects. It was observed that when the object surface is thin and allows some light transmission, the red component appears more saturated on the image sensor. In contrast, highly reflective surfaces or thicker materials that dull the color make it difficult for the HSV filter to distinguish the object from the background.

5.3 Data Interpretation
Linear Relationship: As the robot approaches the target (from *detection_1656* to *detection_1792*), the red pixel count shows a linear increase. This indicates that, under ideal lighting conditions, the camera can provide a reliable proxy for distance estimation.

6. Application Requirement Evaluation and Purchase Recommendation

6.1 Requirement Fulfilment Assessment
The table below summarises how well the TurtleBot 4 meets the core requirements of an autonomous stock-checking application:

Core Requirement	Sensor	Performance Evaluation	Conclusion
Repeatable path navigation	Wheel odometry	Significant systematic bias and large accumulated error	Not satisfied
Obstacle detection & avoidance	2D LiDAR	Highly stable; bias present but can be calibrated	Satisfied
Colour marker recognition	RGB camera	Correct detection logic but low environmental robustness	Partially satisfied

6.2 Final Recommendation: Do Not Purchase in Current State
Rationale: As an out-of-the-box platform, the TurtleBot 4 does not provide sufficient localization accuracy (greater than 15% error) or visual robustness for a stock-checking task. Without substantial algorithmic refinement by a dedicated engineering team, the platform is unlikely to achieve reliable autonomous operation in a real warehouse environment.

7. Required Enhancements
To make the platform suitable for the intended application, the following improvements are necessary:
Odometry Linear Compensation (Software Calibration):
Introduce a data-driven scaling factor within the navigation algorithm. Based on experimental findings, it is recommended to apply a correction of approximately 12%–15% to the odometry linear readings.
URDF Coordinate Calibration:
To correct the systematic positive bias in LiDAR measurements, calibrate the physical offset between the LiDAR sensor and the robot's *base_link* frame within the URDF model.

Appendices A

Pass/Fail criterion: % error < _-10_ %

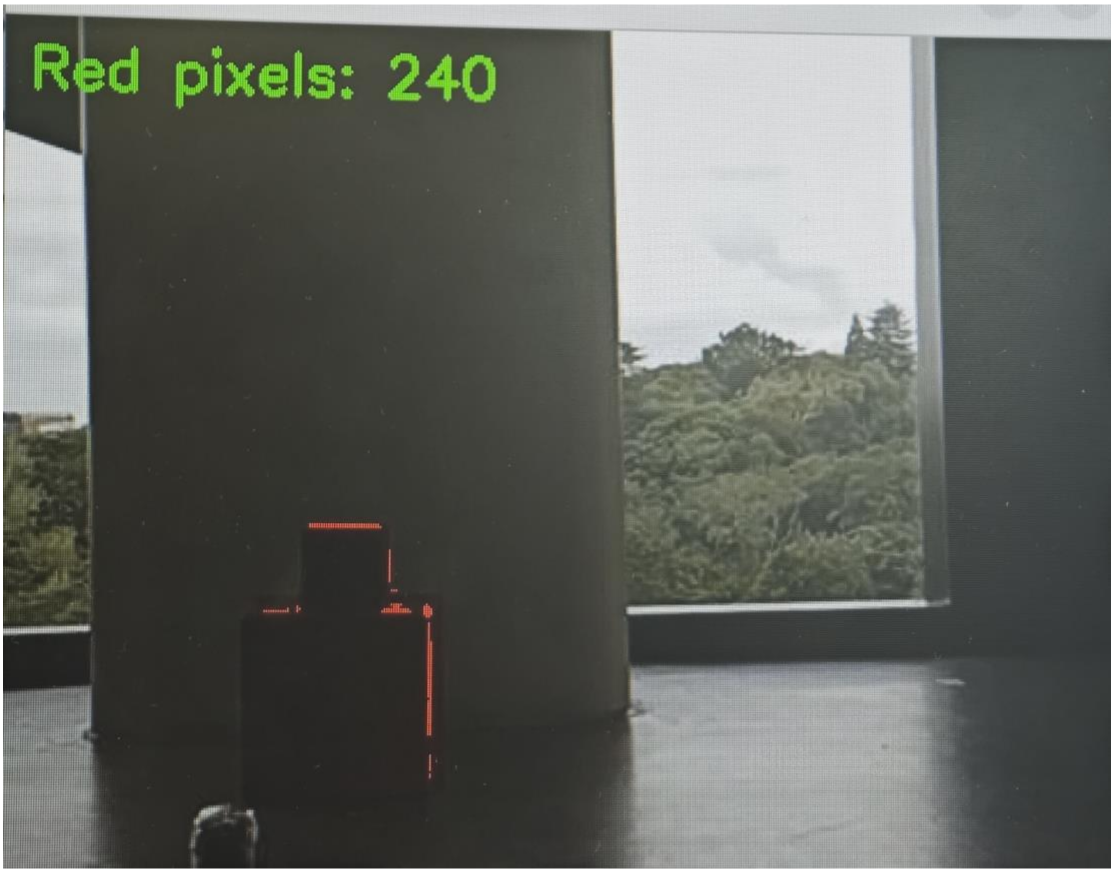
Trial	Target (m)	Odom reported (m)	Physical measurement (m)	Error (m)	% Error	Pass / Fail
1	0.5	0.4244	0.43	-0.0756	-15.12	FAIL
2	0.5	0.3354	0.34	-0.1646	-32.92	FAIL
3	0.5	0.3987	0.40	-0.1013	-20.26	FAIL
4	0.5	0.406	0.40	-0.094	-18.8	FAIL
5	0.5	0.4316	0.44	-0.0684	-13.68	FAIL
6	1.0	0.8903	0.90	-0.1097	-10.97	FAIL
7	1.0	0.7473	0.75	-0.2527	-25.27	FAIL
8	1.0	0.8356	0.84	-0.1644	-16.44	FAIL
9	1.0	0.8367	0.84	-0.1633	-16.33	FAIL
10	1.0	0.9123	0.90	-0.0877	-8.77	PASS
11	1.5	1.3923	1.40	-0.1077	-7.18	PASS
12	1.5	1.408	1.40	-0.092	-6.13	PASS
13	1.5	1.3158	1.33	-0.1842	-12.28	FAIL
14	1.5	1.3827	1.40	-0.1173	-7.82	PASS
15	1.5	1.4143	1.43	-0.0857	-5.71	PASS
16	2.0	1.7822	1.80	-0.2178	-10.89	FAIL
17	2.0	1.9022	1.90	-0.0978	-4.89	PASS
18	2.0	1.8164	1.84	-0.1836	-9.18	PASS
19	2.0	1.8309	1.83	-0.1691	-8.46	PASS
20	2.0	1.8882	1.89	-0.1118	-5.59	PASS

Appendices B: Summary of LiDAR ranging performance statistics

Ground truth(m)	Std Dev	Mean(m)	Error (%)
0.5m	50	0.518m	+3.60%
1.0m	50	1.167m	+16.70%
2.0m	50	2.152m	+7.60%

Appendices C： Images of objects with thinner surfaces versus objects with thicker surfaces under different lighting conditions

This is the condition light is poor.



This is the condition light is strong, but the surface of red object up is thick, red object down is thin.

Appendices D

Trial	Path Len (m)	Closing Error (m)	Final X (m)	Final Y (m)	Heading (°)	Status
T1	1.0000	0.1172	-0.1051	0.0520	-3.0600	FAIL
T2	1.0000	0.0784	-0.0784	-0.0024	-4.0200	PASS
T3	1.0000	0.0743	-0.0685	0.0289	2.8700	PASS

Summary Statistics

Metric	Value
Mean closing error (m)	0.0900
Std Dev (m)	0.0193
Min closing error (m)	0.0743
Max closing error (m)	0.1172
Pass rate (< 0.1m)	2/3

Appendices E: *Snapshot3.png*